

Lemma 1

Let $A = (a_{ij})$ be an $n \times n$ matrix. For numbers $b_1, b_2, \dots, b_n$ Consider the matrices

$$B = A + \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ b_{i1} & b_{i2} & \dots & b_{in} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}, \quad C = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{i1} & b_{i2} & \dots & b_{in} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

B is the same as matrix A except i^{th} row is
 $a_{i1} + b_{i1} \quad a_{i2} + b_{i2} \quad \dots \quad a_{in} + b_{in}$

i^{th} row of A replaced by
 the row $b_1, b_2, \dots, b_n$

$$\text{Then } \det(B) = \det(A) + \det(C).$$

(Idea!)

Proof: We know that every term in $\det(B)$ is a product of n entries from B and that an entry from i^{th} row appears only once.

$$\text{Thus } \det(B) = \dots + (a_{ij} + b_{ij}) \cdot X + \dots,$$

where X is a product of entries of from other rows.

$$\text{So } \det(B) = \dots + a_{ij} X + b_{ij} X + \dots,$$

Each term splits this way and we obtain

$$\det(B) = \det(A) + \det(C)$$

Lemma ② If two adjacent rows of a $n \times n$ matrix are the same then its determinant is 0 ($n \geq 2$)

Proof The proof will use an induction on n .

Step 1 $n=2$. We have $\det \begin{pmatrix} a_{11} & a_{12} \\ a_{11} & a_{12} \end{pmatrix} = a_{11}a_{12} - a_{11}a_{12} = 0$

Step 2 Assume that the lemma is true for $(n-1) \times (n-1)$ matrices

* If the identical rows are i and $i+1$, where $i \geq 2$ then the lemma can be easily proved using the induction hypothesis!

(Why?)

Suppose that the adjacent rows are the 1st and 2nd rows

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{pmatrix}$$

We know,

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + a_{13} \det(A_{13}) - \dots + (-1)^{n+1} a_{1n} \det(A_{1n})$$

Observe that

$$a_{11} \det(A_{11}) = a_{11} \left[a_{12} \det(B_{12}) - a_{13} \det(B_{13}) + \dots + (-1)^{n-2} a_{1n} \det(B_{1n}) \right]$$

$$-a_{12} \det(A_{12}) = -a_{12} \left[a_{11} \det(B_{12}) - a_{13} \det(*) + \dots + (-1)^{n-2} a_{1n} \det(\bullet) \right]$$

$$a_{13} \det(A_{13}) = a_{13} \left[a_{11} \det(B_{13}) - a_{12} \det(*) + \dots \right]$$

⋮

$$(-1)^{n-1} a_{1n} \det(A_{1n}) = (-1)^{n-1} a_{1n} \left[a_{11} \det(B_{1n}) - a_{12} \det(\bullet) + \dots \right]$$

Thus $\det(A) = 0$

Lemma 3 If B is the matrix obtained by switching two adjacent rows of a matrix A then $\det(A) = -\det(B)$

Proof

Write

$$A = \begin{pmatrix} R_1 \\ \vdots \\ R_i \\ R_{i+1} \\ \vdots \\ R_n \end{pmatrix}$$

R_i - i^{th} row of A .

$$B = \begin{pmatrix} R_1 \\ \vdots \\ R_{i+1} \\ R_i \\ \vdots \\ R_n \end{pmatrix}$$

and let

$$C = \begin{pmatrix} R_1 \\ \vdots \\ R_i + R_{i+1} \\ R_i + R_{i+1} \\ \vdots \\ R_n \end{pmatrix}$$

Then, $\det(C) = 0$. Also, we know that

$$0 = \det(C) = \det \begin{pmatrix} R_1 \\ \vdots \\ R_i \\ R_i + R_{i+1} \\ \vdots \\ R_n \end{pmatrix} + \det \begin{pmatrix} R_1 \\ \vdots \\ R_{i+1} \\ R_i + R_{i+1} \\ \vdots \\ R_n \end{pmatrix}$$

$$\Rightarrow = \det \begin{pmatrix} R_1 \\ \vdots \\ R_i \\ R_i \\ \vdots \\ R_n \end{pmatrix} + \det \begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_n \end{pmatrix} + \det \begin{pmatrix} R_1 \\ \vdots \\ R_{i+1} \\ R_i \\ \vdots \\ R_n \end{pmatrix} + \det \begin{pmatrix} R_1 \\ \vdots \\ R_{i+1} \\ R_{i+1} \\ \vdots \\ R_n \end{pmatrix}$$

$$0 = 0 + \det(A) + \det(B) + 0$$

Thus, we have shown that $\det(A) = -\det(B)$.

Proposition 4

If i^{th} and j^{th} rows of a matrix A are identical

then $\det(A) = 0$

Proof. Let $i < j$. Then, we may switch rows of A as many times needed and obtain a matrix B such that i and $i+1$ rows are identical. Now,

$$\det(A) = (-1)^r \det(B) = (-1)^r \cdot 0 = 0$$

Proposition 5 If B is the matrix obtained by switching any two rows of a matrix A then $\det(A) = -\det(B)$

Proof (Hint) In order to switch i^{th} row with j^{th} row, where $i < j$, we need $2(j-i)-1$ switches of adjacent rows. Since $2(j-i)-1$ is an odd number and

$$\det(A) = (-1)^{2(j-i)-1} \det(B),$$

we obtain that

$$\det(A) = -\det(B)$$

Row operations. Let A be an $n \times n$ matrix

1. If $R_i \rightarrow R_i + \lambda R_j$ then for $B =$

$$\boxed{\det(B) = \det(A)}$$

This is because, $\det(B) = \det(A) + \det$

$$\begin{aligned} &= \det(A) + \lambda \det \begin{pmatrix} R_1 \\ \vdots \\ R_i \\ \vdots \\ R_j + \lambda R_i \\ \vdots \\ R_j \\ \vdots \\ R_n \end{pmatrix} \\ &= \det(A) + \lambda \det \begin{pmatrix} R_1 \\ \vdots \\ R_i \\ \vdots \\ R_j \\ \vdots \\ R_j + \lambda R_i \\ \vdots \\ R_n \end{pmatrix} \\ &= \det(A) + 0 = \det(A). \end{aligned}$$

2. If $R_i \longrightarrow \lambda R_i$ then for $B = \begin{pmatrix} R_1 \\ \vdots \\ \lambda R_i \\ \vdots \\ R_n \end{pmatrix}$,

$$\det(B) = \lambda \det(A)$$

3. If $R_i \longleftrightarrow R_j$ then for $B = \begin{pmatrix} R_1 \\ \vdots \\ R_j \\ \vdots \\ R_i \\ \vdots \\ R_n \end{pmatrix}$,

$$\det(A) = -\det(B).$$